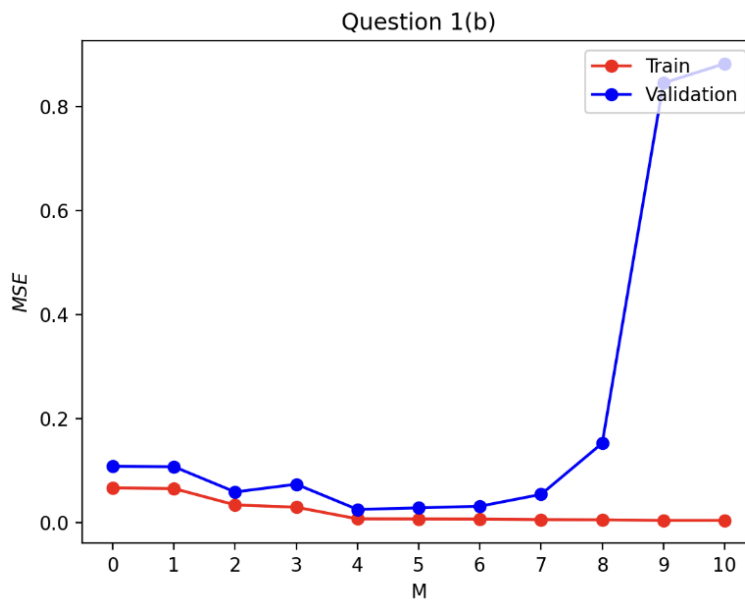


1.a.

```
27 def generate_polynomial_features(X, M):
28     """
29     Create a polynomial feature mapping from input examples. Each element x
30     in X is mapped to an (M+1)-dimensional polynomial feature vector
31     i.e.  $x \rightarrow [1, x, x^2, \dots, x^M]$ .
32
33     Args:
34         X: np.array, shape (n, 1). Each row is one instance.
35         M: a non-negative integer
36
37     Returns:
38         Phi: np.array, shape (n, M+1)
39     """
40     n, _ = X.shape
41     Phi = np.zeros((n, M+1))
42
43     # TODO: (a) - Implement this function
44     arrM = np.arange(M+1)
45     Phi = X ** arrM
46
47     return Phi
48
```

1.b.i.



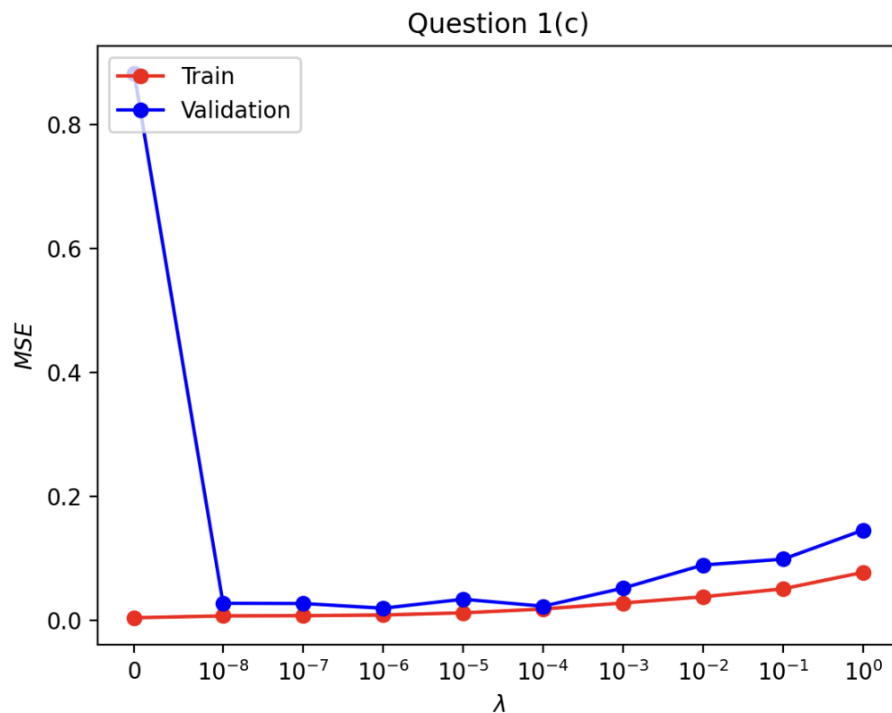
1.b.ii.

The polynomial of degree 4 or 5 best fits the data. Lower degrees (0 and 1) underfit, and higher degrees ($M > 7$) overfit

1.c.i.

```
50 def closed_form_optimization(X, y, reg_param=0):
51     """
52     Implements the closed form solution for least squares regression.
53
54     Args:
55         X: np.array, shape (n, d)
56         y: np.array, shape (n,)
57         reg_param: float, an optional regularization parameter
58
59     Returns:
60         theta: np.array, shape (d,)
61     """
62     # TODO: (c) - Modify implementation to incorporate L2-regularization
63     # setting gradient = 0 ... theta = (XTX/n + lambda*identity)^-1 @ (XTy / n)
64
65     n, d = X.shape
66     # theta = np.linalg.pinv(X.T @ X) @ X.T @ y
67
68     identity = np.eye(d)
69
70     A = (X.T @ X) / n
71     A = A + (reg_param * identity)
72
73     b = (X.T @ y) / n
74
75     # A@theta = b --> theta = A^-1 @ b
76     theta = np.linalg.solve(A, b)
77
78     return theta
```

1.c.ii.



1.c.ii.

The lambda value of $1e-6$ has the lowest validation MSE and has a train MSE very close to zero. We see little change in the train MSE, so we should take what appears to be the smallest validation MSE.

2.a.

Frederick's setup is better because it uses cross-validation within the training set to select hyperparameters, avoiding contaminating the test set. Nathan's approach tunes hyperparameters directly on the test data, leading to overfitting to the test set and an overly optimistic estimate. Therefore, Frederick's method is more likely to reflect true performance on unseen data.

2.b.

The best practice all three would agree on is to use cross-validation on the training data, select hyperparameters on the validation set, and then retrain on the full training set with the best hyperparameters from cross validation. Lastly, the model would be tested and report performance on the untouched test set. This would give the hyperparameters that generalize the best to unseen data and then train a model that has been trained on all the data, leading to the best model and the most accurate performance report.

3.

3a) $K_1(\bar{u}, \bar{v}) = (\bar{u} \cdot \bar{v})$ for $\bar{u}, \bar{v} \in \mathbb{R}^2$ $X = \{x_1, \dots, x_n\} \in \mathbb{R}^2$

Gram matrix $G \in \mathbb{R}^{n \times n} \rightarrow G_{ij} = K_1(x_i, x_j) = x_i^T x_j = x_j^T x_i = G_{ji}$

For any $c \in \mathbb{R}^n \rightarrow c^T G c = c^T X^T X c = \|X^T c\|_2^2 \geq 0$

$\star G$ is symmetric and positive semi-definite for every X , so K_1 is a valid kernel

b) $K_2(\bar{u}, \bar{v}) = (1 + \bar{u} \cdot \bar{v})^2$ on $\mathbb{R}^2$

$\bar{u} = (u_1, u_2) \quad \bar{v} = (v_1, v_2)$

$(1 + \bar{u} \cdot \bar{v})^2 = (1 + u_1 v_1 + u_2 v_2)^2 = 1 + 2u_1 v_1 + 2u_2 v_2 + 2u_1 v_1 u_2 v_2 + u_1^2 v_1^2 + u_2^2 v_2^2$

$\phi(u_1, u_2) = [1, \sqrt{2}u_1, \sqrt{2}u_2, \sqrt{2}u_1 u_2, u_1^2, u_2^2]$
 $\phi(v_1, v_2) = [1, \sqrt{2}v_1, \sqrt{2}v_2, \sqrt{2}v_1 v_2, v_1^2, v_2^2]$ + same for v's

$K_2(u, v) = \phi(u) \cdot \phi(v)$

c) $K_3(\bar{u}, \bar{v}) = 4(\bar{u} \cdot \bar{v}) + (1 + \bar{u} \cdot \bar{v})^2$ on $\mathbb{R}^2$

• from part a, $\bar{u} \cdot \bar{v}$ is a valid kernel

↳ if K is a kernel and $a \geq 0$, the aK is a kernel, so $4K_1$ is a kernel

• from part b, $(1 + \bar{u} \cdot \bar{v})^2$ is a valid kernel

• for $K_3 = 4K_1 + K_2$, the sum of kernels is a kernel

• thus, K_3 is a valid kernel

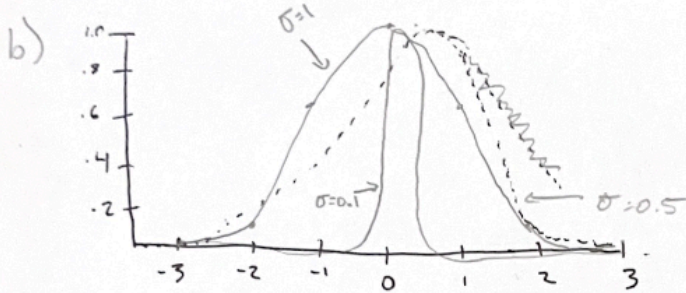
$$4a) \quad k(\bar{u}, \bar{v}) = e^{(-\frac{1}{2\sigma}(\|\bar{u} - \bar{v}\|^2))}$$

$$i. \quad k(\bar{u}, \bar{v}) = e^{(-1)(0^2)} = 1$$

$$ii. \quad k(\bar{u}, \bar{v}) = e^{(-1)(-1)^2} = e^{-1}$$

$$iii. \quad k(\bar{u}, \bar{v}) = e^{(-1)((-2)^2)} = e^{-4}$$

RBF kernel value decreases as points get further and further apart from each other. 1 is maximum similarity. Lower bound approaches 0.



→ smaller σ , sharper curve
→ larger σ , flatter curve

→ A smaller σ value only considers very close points as similar. Larger σ values treat distant points as somewhat similar.

$$\sigma = 1$$

-3:	$e^{-(\frac{9}{2})} = e^{-4.5} \approx 0.011$
-2:	$e^{-(\frac{4}{2})} = e^{-2} = 0.135$
-1:	$e^{-(\frac{1}{2})} = e^{-0.5} = 0.61$
0:	$e^0 = 1$
1:	$e^{-(\frac{1}{2})} = e^{-0.5} = 0.61$
2:	$e^{-2} = 0.135$
3:	$e^{-4.5} \approx 0.011$

$$\sigma = 0.1$$

-3:	$e^{-(\frac{9}{0.2})} = 0$
-2:	$e^{-(\frac{4}{0.2})} = 0$
-1:	$e^{-(\frac{1}{0.2})} = 0$
0:	$e^0 = 1$
1:	$e^{-(\frac{1}{0.2})} = 0$
2:	$e^{-(\frac{4}{0.2})} = 0$
3:	$e^{-(\frac{9}{0.2})} = 0$

c) The RBF kernels notion of similarity decides how much influence a point has. For a sensitive metric (small σ value), the decision boundary would be highly complex and overfit the training data. On the flip side, a less sensitive metric (larger σ value) might have a looser decision boundary and underfit the training data.

5.

5 a)

$$\bar{\theta}_c = \sum_{i=1}^n a_{i,c} \phi(\bar{x}^{(i)})$$

$$\bar{\theta}_c^T \phi(x) = \sum_{i=1}^n a_{i,c} \underbrace{\phi(\bar{x}^{(i)})^T \phi(\bar{x})}_{K(\bar{x}^{(i)}, \bar{x})} = \sum_{i=1}^n a_{i,c} K(\bar{x}^{(i)}, \bar{x})$$

So, $\hat{y} = \underset{c \in \{1, \dots, m\}}{\operatorname{argmax}} \sum_{i=1}^n a_{i,c} K(\bar{x}^{(i)}, \bar{x})$

b) Initialization: for $c=1, \dots, m$: $\bar{\theta}_c^{(0)} = \bar{0}$, $k=0$
Repeat until convergence:

for $i=1, \dots, n$:

$$\hat{y}^{(k)} = \underset{c}{\operatorname{argmax}} \sum_{i=1}^n a_{i,c} K(\bar{x}^{(i)}, \bar{x})$$

if $\hat{y}^{(k)} \neq y^{(i)}$:

$$a_{i, y^{(i)}} = a_{i, y^{(i)}} + 1$$

$$a_{i, \hat{y}} = a_{i, \hat{y}} - 1$$

$k \leftarrow k+1$

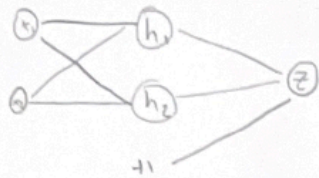
c) $h(\bar{x}) = \underset{c}{\operatorname{argmax}} \sum_{i=1}^n a_{i,c} K(\bar{x}^{(i)}, \bar{x})$

~~***~~ revisit

@240

6.

6.

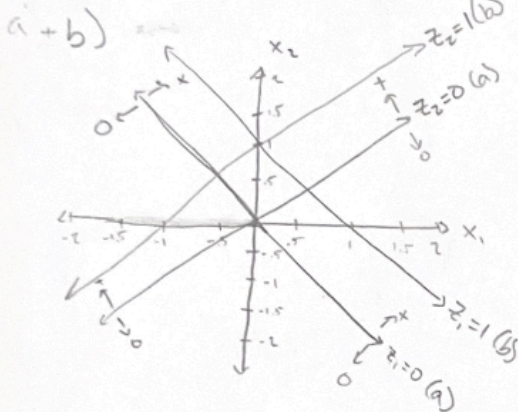


$$h(\bar{x}; \bar{\theta}) = z = v_0 + h_1 + h_2 + v_0$$

$$z_1 = w_{11}x_1 + w_{21}x_2 \quad h_1 = \max\{0, z_1\}$$

$$z_2 = w_{12}x_1 + w_{22}x_2 \quad h_2 = \max\{0, z_2\}$$

$$\begin{aligned} w_{11} &= 1 & v_1 &= 1 \\ w_{12} &= -1 & v_2 &= -1 \\ w_{21} &= 1 & v_0 &= 1 \\ w_{22} &= 1 & & \end{aligned}$$



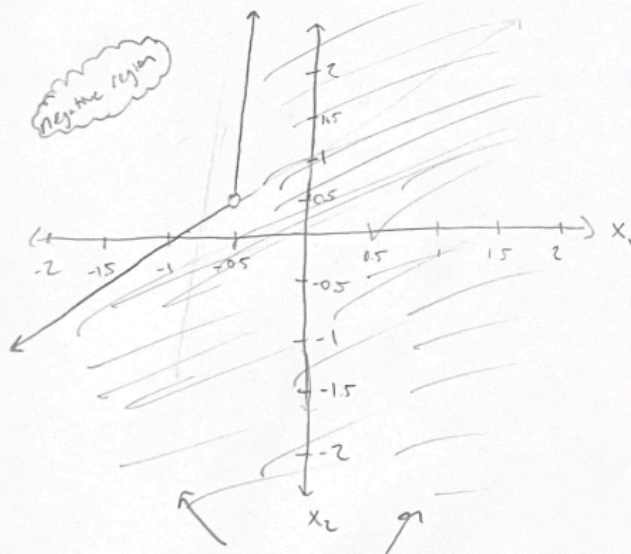
$$z_1 = x_1 + x_2 \rightarrow h_1: x_1 + x_2 \leq 0 \rightarrow x_1 = -x_2$$

$$h_1 = \begin{cases} m=0 & x_1 + x_2 \leq 0 \\ m=1 & x_1 + x_2 > 0 \end{cases}$$

$$z_2 = -x_1 + x_2 \rightarrow h_2: -x_1 + x_2 \leq 0 \rightarrow x_1 = x_2$$

$$h_2 = \begin{cases} m=0 & -x_1 > x_2 \\ m=1 & -x_1 \leq x_2 \end{cases}$$

c)



shaded region is POSITIVE

Case 1: both ReLUs off: 1+0

$$x_1 + x_2 \leq 0 \quad -x_1 + x_2 \leq 0 \rightarrow x_1 \geq 0$$

$h_1 = 1$, always positive, no boundary

Case 2: both ReLUs on: 1+1

$$x_1 + x_2 > 0 \quad -x_1 + x_2 > 0 \rightarrow x_1 < -1/2$$

$$h = (x_1 + x_2) - (-x_1 + x_2) + 1 = 2x_1 + 1 = 0$$

$$z_{x_1} = 1$$

$$x_1 = -1/2$$

valid when $x_2 > |x_1|$

$$\text{so } x_2 > 0.5$$

Case 3: h_1 on, h_2 off

$$x_1 + x_2 > 0 \quad -x_1 + x_2 \leq 0$$

$$h = x_1 + x_2 + 1 = 0, \text{ smallest } h \text{ can be is } 1$$

always > 0

always pos, no boundary

Case 4: h_1 off, h_2 on

$$x_1 + x_2 \leq 0 \quad -x_1 + x_2 > 0 \rightarrow x_2 > x_1 \text{ and}$$

$$h = 0 - (-x_1 + x_2) + 1 = 0$$

$$x_2 = x_1 + 1$$

$$x_1 + x_2 \leq 0,$$

$$\text{so } x_1 \leq -1/2$$

7.a.i.

Because `df_out.ffmpeg()` only propagates forward, if a variables' first windows have no measurements, their mean will be `np.nan` and moving forward those missing values will remain `np.nan`. Also, if a patient has no readings at all for a time series variable, then every window's mean is `nan`. Forward fill can't invent a value, so the whole column for that variable remains `np.nan`.

7.a.ii.

Resampling creates a regular, fixed-length sequence so a standard LSTM can process the data with consistent time steps and batching. This ensures that each patient's record has the same number of windows and variables which allows the LSTM to uniformly model changes across patients.

7.b.

`weight_ih` - the learnable input-hidden weights, of shape $(4 \times \text{hidden_size}, \text{input_size})$
- $4 * 128 * 70$

`weight_hh` (torch.Tensor) - the learnable hidden-hidden weights, of shape $(4 \times \text{hidden_size}, \text{hidden_size})$ - $4 * 128 * 128$

`bias_ih` - the learnable input-hidden bias, of shape $(4 \times \text{hidden_size})$
- $4 * 128$

`bias_hh` - the learnable hidden-hidden bias, of shape $(4 \times \text{hidden_size})$
- $4 * 128$

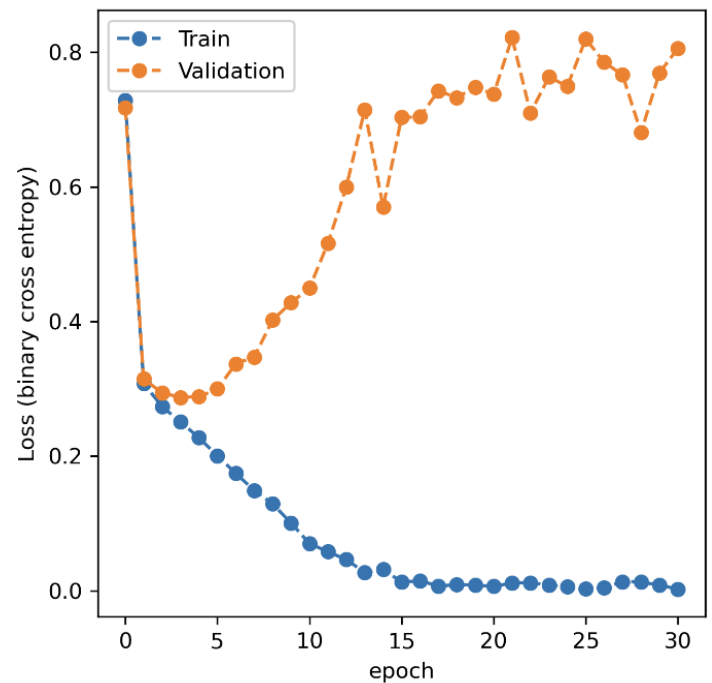
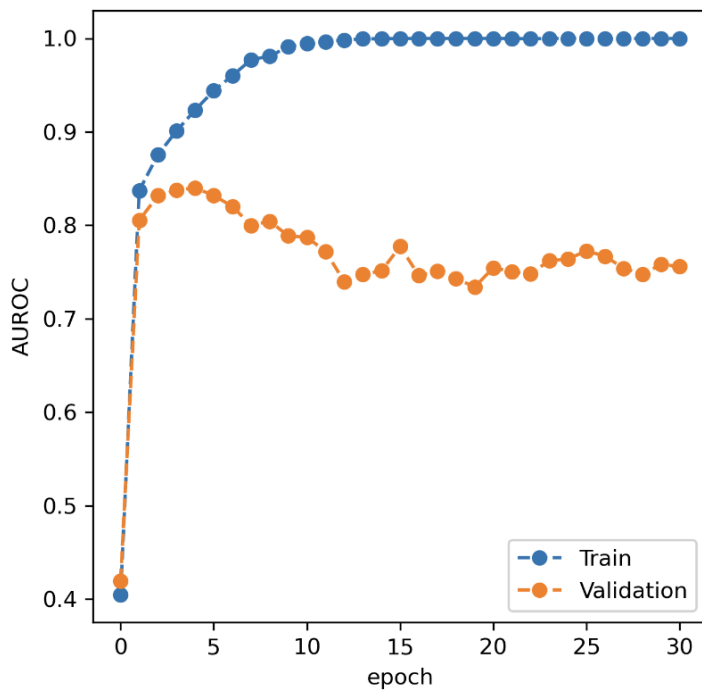
$\text{total} = 35840 + 65536 + 512 + 512 + (128 \text{ output weights} + 1 \text{ bias from linear layer}) = 102,400 + 129 = 102,529 \text{ total parameters}$

7.c.

```
1 import torch
2 import torch.nn as nn
3
4 class RNN(nn.Module):
5     def __init__(self, input_size, hidden_size, output_size):
6         """
7         Initializes the RNN model. Defines LSTMCell and fully connected layer.
8
9         Args:
10            input_size: Dimension of the input features.
11            hidden_size: Dimension of the hidden state in the LSTM.
12            output_size: Dimension of the output layer.
13        """
14
15        super().__init__()
16        # Define the parameters of the RNN
17        self.hidden_size = hidden_size
18
19        # TODO: Uncomment the next two lines and replace the ???s
20        self.lstm = nn.LSTMCell(input_size, hidden_size)
21        self.fc = nn.Linear(hidden_size, output_size)
22
23        self.init_weights()
24
25    def init_weights(self):
26        """
27        Initializes the weights of LSTMCell and fully connected layer.
28        Biases are initialized to zero and weights using Xavier uniform initialization.
29        """
30        # TODO: Initialize the weights of the RNN
31
32        # NOTE:: Use "for name, param in self.lstm.named_parameters():" to
33        # loop over the parameters of the LSTMCell. For the fully connected
34        # layer, use xavier_uniform_ to initialize the weights and constant_ to
35        # initialize the bias to zero.
36
37        for name, param in self.lstm.named_parameters():
38            if 'weight' in name:
39                nn.init.xavier_uniform_(param.data)
40            elif 'bias' in name:
41                nn.init.constant_(param.data, 0.0)
42
43        nn.init.xavier_uniform_(self.fc.weight.data)
44        nn.init.constant_(self.fc.bias.data, 0.0)
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7.d.

It looks like we start over fitting at epoch 5 when we see the validation AUROC score decline for the first time



7.e.

=> Successfully restored checkpoint (trained for 4 epochs)

Test loss : 0.34897807240486145

Test AUROC: 0.820764163372859

7.f.i.

If Xavier initialization is replaced with constant initialization, the model will converge much more slowly or fail to converge because all neurons start identically and gradients cannot break symmetry.

7.f.ii.

Another hyperparameter we could play around with would be batch size. Lets decrease ours a little bit to 32 instead of 64 and check out performance...

7.f.iii.

You could incorporate static features by concatenating them with the LSTM's final hidden state before the fully connected layer, so the model jointly uses time-varying and time-invariant information.

=> Successfully restored checkpoint (trained for 4 epochs)

Test loss : 0.36151470031057087

Test AUROC: 0.8145981554677207

We see a little better loss but slightly worse AUROC with the reduced batch size. Training also took much longer which was expected.

7.f.iv.

LSTMs use two states + four gates to separately control long and short-term memory, while GRUs use one state and two gates. So, replacing LSTMs with GRUs would make the model smaller and faster to train but possibly less effective at capturing very long-term data. Over the summer in my internship, I used GRUs to classify malicious vs benign LLM prompts for the AI Governance team at Lumen and it worked well!

8.a.

attention scores for the word 'ML': [-0.3695 0.1501 -0.4157]

The largest score of 0.1501 means that the model finds the word "like" to be the most similar to "ML".

8.b.

attention weights for the word 'ML': [0.275 0.462 0.263]

context-aware representation (context vector) for the word 'ML': [-0.06256849 0.18003837 0.56139104]

8.c.

Multi-head attention allows the model to attend to different parts of a sentence at once which improves performance by capturing more diverse contextual relationships compared to a single attention head.

8.d.

The softmax function normalizes the attention scores into probabilities that sum to 1 and makes sure the model assigns meaningful relative importance to different words.

8.e.

Scaling by $\sqrt{d_k}$ prevents dot products from becoming too large in high-dimensional spaces and keeps the softmax function from saturating.

8.f.

Attention captures long-range dependencies by directly linking all words in a sequence. Contrary to RNNs, it doesn't depend on step-by-step propagation.